

# Efficient and Equitable Taxation

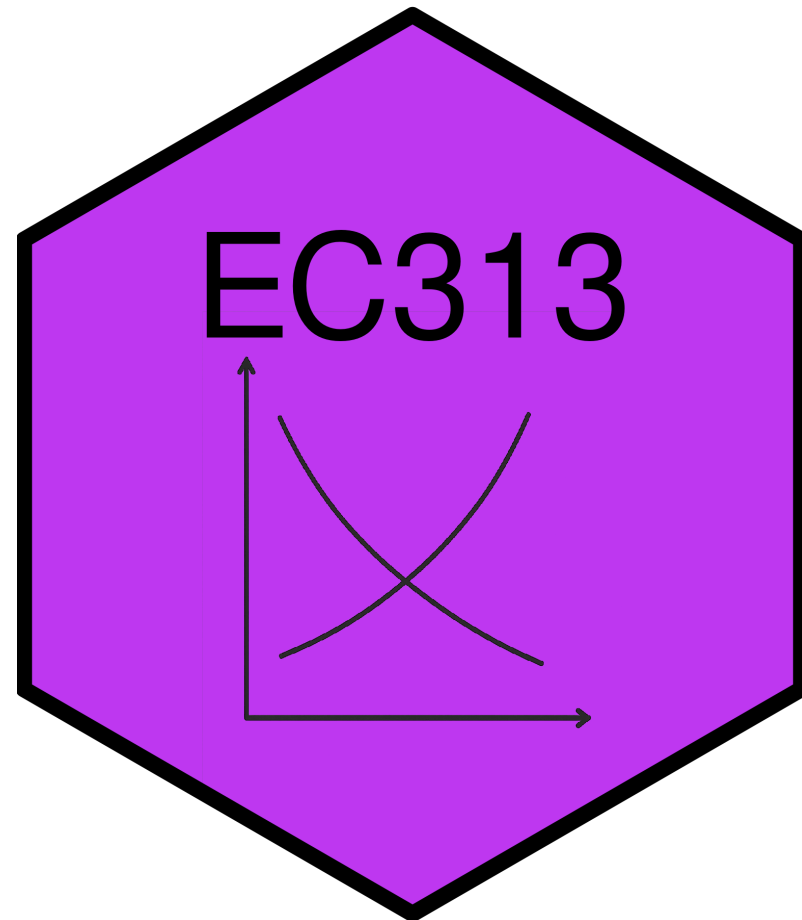
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EC313 - Public Economics: Taxation

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# Goals of This Section

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In this section we will:

- Lay out model for optimal taxation
- Derive the Ramsey Rule for optimal commodity taxation
- Discuss optimal user fees
- Discuss optimal income taxation
- Discuss political economy and time inconsistency
- Discuss other criteria for tax design

# Optimal Taxation

# Introduction

- As we have discussed, government raise taxes mainly to raise **revenue**
- But taxes involve costs
  - The direct cost of the revenue raised
  - Additional costs from distortions in behaviour (**excess burden**)
- Given that revenue needs to be raised, how should taxes be designed to minimize costs?
- **Optimal taxation** addresses this question

# Model

- Question of optimal taxation is answered with a model
- A representative citizen consumes two goods,  $X$  and  $Y$ 
  - Prices of these goods are  $P_X$  and  $P_Y$
- Person can also work and earn wage  $w$ 
  - Person has  $T$  hours available for work ( $h$ ) or leisure ( $L$ )
  - These are the only two uses of time, so  $T = h + L$
- Their budget constraint in this context is

$$w(T - L) = P_X X + P_Y Y$$

- Constraint says that income is split between spending on goods  $X$  and  $Y$

# Model

- If you rearrange the budget constraint, you can write it as

$$wT = P_X X + P_Y Y + wL$$

- Shows that **full income** is split between spending on goods  $X$  and  $Y$  and spending on leisure
  - Full income is  $wT$  because if all time was worked, income would be  $wT$
  - Price of leisure is  $w$  because each hour of leisure foregoes  $w$  in earnings
- Suppose we tax goods  $X$ ,  $Y$ , and  $L$  at rate  $t$
- Then the budget constraint becomes

$$wT = (1 + t)P_X X + (1 + t)P_Y Y + (1 + t)wL$$

# Model

- Rearranging gives

$$wT = (1 + t)(P_X X + P_Y Y + wL)$$

$$\frac{wT}{1 + t} = P_X X + P_Y Y + wL$$

## Key Insight

A tax on all goods and leisure is equivalent to a reduction in full income by a factor of  $\frac{1}{1+t}$

- We saw this in one of the practice questions

# Model

## ! Why Taxing Everything Equally Is Non-Distortionary

If we could tax all goods and leisure at the same rate, we would **not** distort behaviour because **relative prices are unchanged**. The commodity taxes are equivalent to a **lump-sum tax** that reduces income.

- But in practice, we **cannot** tax all goods and leisure at the same rate
  - We cannot tax leisure directly
  - Can only tax  $X$  and  $Y$  in this setup
- Taxing some goods ( $X$  and  $Y$ ) but not others ( $L$ ) distorts relative prices and leads to **excess burden**

# Ramsey Rule

- Question then becomes: If we cannot tax all goods at the same rate, how should we tax them?
- One option is **neutral taxation**: taxing  $X$  and  $Y$  at the same rate
  - This is **simple and easy to administer**
  - But it still **distorts behaviour** and creates excess burden

## ! The Ramsey Rule

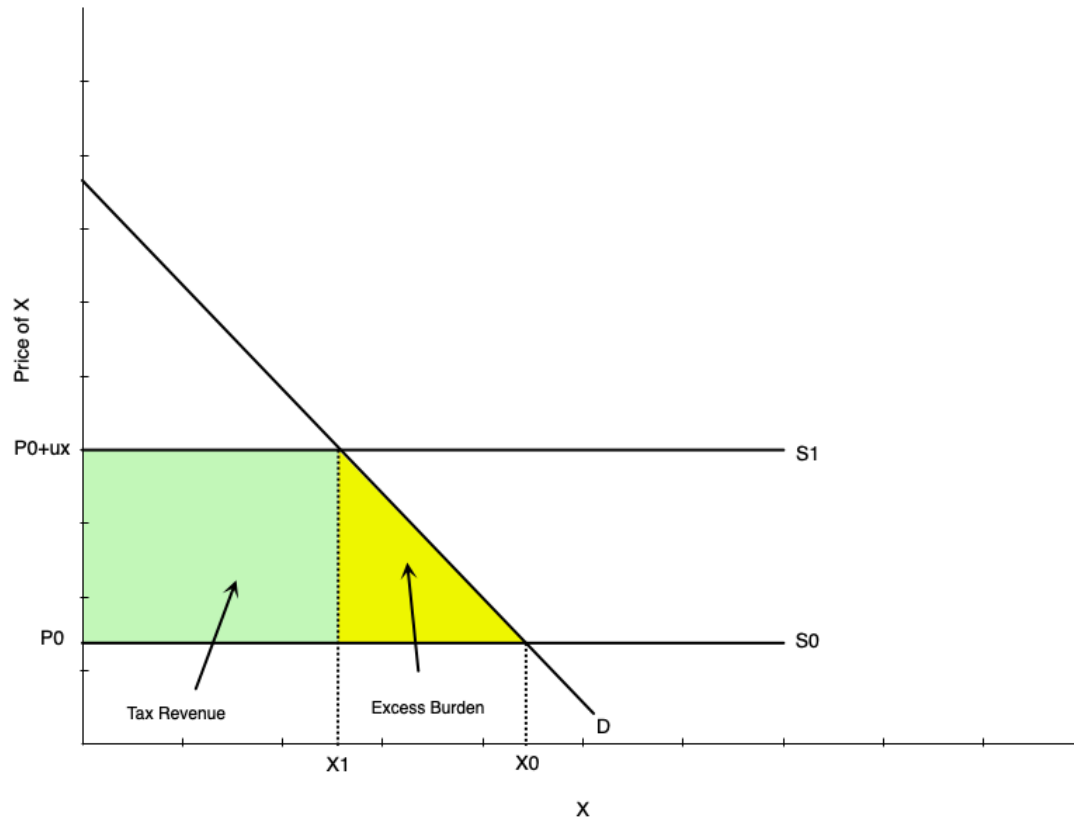
Tax goods so that the **percent reduction in quantity demanded is the same for all goods**. Equivalently, tax goods so that the marginal excess burden of the last dollar of revenue raised is the same for all goods.

- The next few slides establish this result

# Ramsey Rule

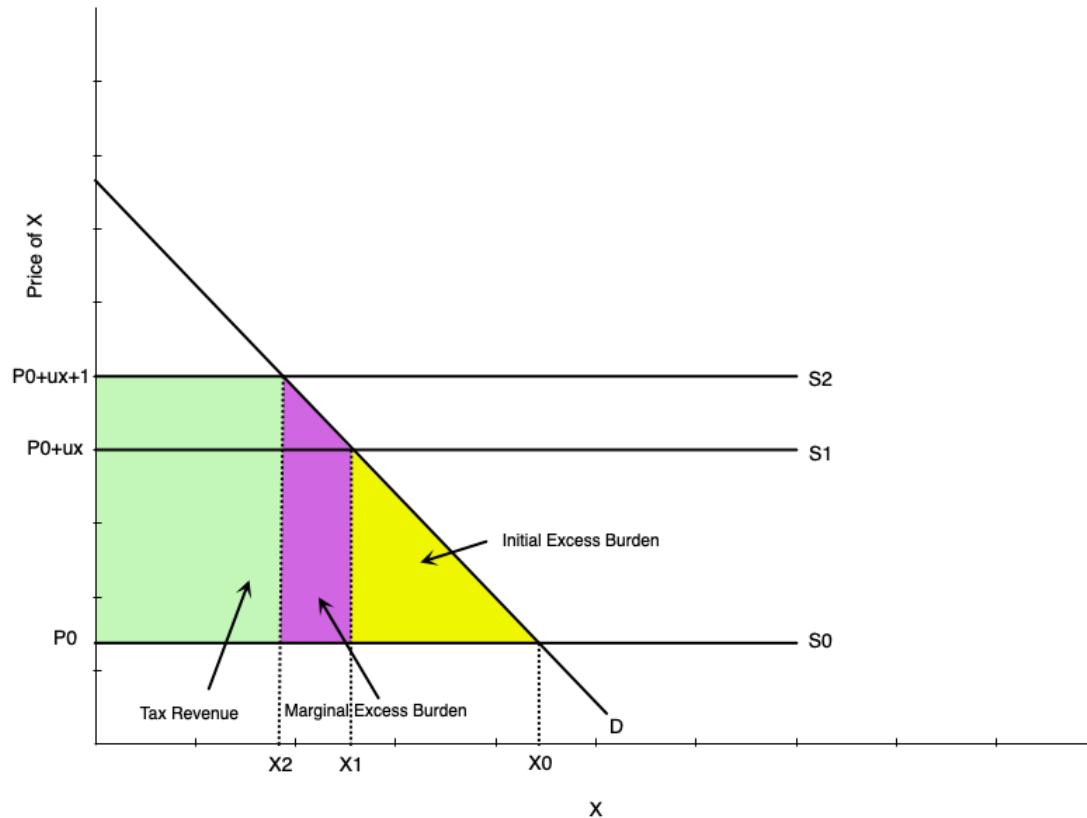
- Consider again two goods,  $X$  and  $Y$ , with prices  $P_X$  and  $P_Y$
- Pretend that goods are neither substitutes nor complements
  - Simplifies the analysis because changes in their prices do not affect each other's demand
  - Results still hold for substitutes and complements, but analysis is more complex
- Also pretend the supply curve is horizontal
  - Consumers face the entire economic incidence of the tax
  - Again simplifies the analysis because we do not have to consider supply side effects
- Consider a unit tax  $u_X$  on good  $X$

# Ramsey Rule



- Consumer initially consumes  $X_0$  at price  $P_0$
- After tax  $u_X$  is imposed, price rises to  $P_0 + u_X$
- Consumer reduces consumption to  $X_1$
- As we have learned this involves an **excess burden**
  - Equals area of yellow triangle
  - Represents the loss in consumer surplus

# Ramsey Rule



- Now imagine raising the tax further from  $u_X$  to  $u_X + 1$
- The additional excess burden created is the **marginal excess burden**
  - Equals area of purple triangle
- Part of that excess burden was **tax revenue** (purple rectangle)
- Other part is additional **loss in consumer surplus** (purple triangle)

# Ramsey Rule

- The **marginal excess burden** of the tax on good  $X$  is the purple area
  - Equals area of the purple triangle plus the purple rectangle
- To compute that mathematically, set  $\Delta x = X_1 - X_2$
- The total area is then

$$\left(\frac{1}{2}\Delta x \times 1\right) + (\Delta x u_X)$$

- As noted in the textbook, if we pretend that  $\frac{1}{2}\Delta x$  is very small, we can ignore it
  - So marginal excess burden is approximately  $\Delta x u_X$

# Ramsey Rule

- Going one step further, note that the slope of the demand curve is

$$\frac{u_X}{\Delta X} = \frac{1}{\Delta x}$$

- $\Delta X = X_0 - X_1$ , the initial change in quantity demanded from the initial tax
- Both represent the rise over the run
- Rearranging gives

$$\Delta x u_X = \Delta X$$

- The marginal excess burden of the tax on good  $X$  is approximately  $\Delta X$ 
  - The change in quantity demanded from the initial tax

# Ramsey Rule

- What about the change in revenue?
- Initially revenue was larger green rectangle plus purple rectangle
- After, it is larger green rectangle plus smaller green rectangle
- The **marginal tax revenue** is therefore the smaller green rectangle minus the purple rectangle

$$(X_2 \times 1) - (\Delta x u_X)$$

- We previously saw that  $\Delta x u_X = \Delta X$
- We also know that  $X_2 = X_1 - \Delta x$
- Subbing in gives us

$$X_1 - \Delta x - \Delta X$$

# Ramsey Rule

- IF we pretend that  $\Delta x$  is very small, we can ignore it
  - So marginal tax revenue is approximately  $X_1 - \Delta X$
- The marginal excess burden *per dollar of additional revenue* is

$$\frac{\Delta X}{X_1 - \Delta X}$$

- If we repeat this exact exercise for good  $Y$ , we find that the marginal excess burden per dollar of additional revenue is

$$\frac{\Delta Y}{Y_1 - \Delta Y}$$

# Ramsey Rule

- If we set them equal

$$\frac{\Delta X}{X_1 - \Delta X} = \frac{\Delta Y}{Y_1 - \Delta Y}$$

- Which simplifies to the [Ramsey Rule](#)

$$\frac{\Delta X}{X_1} = \frac{\Delta Y}{Y_1}$$

- If you multiply both sides by 100 to express this as a percentage, it says percent change in quantity demanded of  $X$  equals percent change in quantity demanded of  $Y$

# Ramsey Rule with Elasticities

- Economists enjoy expressing things in terms of **elasticities**
- The price elasticity of demand is

$$\eta_X = \frac{\Delta X}{\Delta P_X} \times \frac{P_X}{X}$$

- Rearrange that to get

$$\frac{\Delta X}{X} = \eta_X \times \frac{\Delta P_X}{P_X}$$

# Ramsey Rule with Elasticities

- In this case, consider instead an ad valorem tax that makes the price rise to  $P_X(1 + t_X)$
- The numerator of this fraction is then  $\Delta P_X = P_X(1 + t_X) - P_X = P_X t_X$
- Substituting into the equation on the previous slide gives

$$\frac{\Delta X}{X} = \eta_X \times t_X$$

# Ramsey Rule with Elasticities

- The Ramsey Rule then becomes

$$\eta_X t_X = \eta_Y t_Y$$

- Equivalently

$$\frac{t_X}{t_Y} = \frac{\eta_Y}{\eta_X}$$

# Ramsey Rule with Elasticities

## ! The Inverse Elasticity Rule

The optimal tax rate on a good is **inversely proportional** to its price elasticity of demand. If good  $Y$  has a high elasticity relative to good  $X$ , then  $t_X$  should be higher than  $t_Y$ .

- Why?
  - Because a tax on a good with a **high elasticity** will cause a large reduction in quantity demanded
  - This creates a **large excess burden**
  - So to minimize excess burden, tax goods with high elasticities **less**

## Think About It

The inverse elasticity rule says we should tax goods with inelastic demand more heavily. Demand for food and medicine tends to be very inelastic, while demand for luxury goods like yachts and designer handbags tends to be more elastic. What does the Ramsey Rule imply about how we should tax these goods, and does that sit well with you? What might be missing from a purely efficiency-based approach to tax design?

# Corlett-Hague Rule

- Recall that we could not tax leisure directly
  - And therefore could not apply an efficient (lump-sum equivalent) tax
  - This created the excess burden

## The Corlett-Hague Rule

Approximate a leisure tax by taxing goods that are **complements** (used together with) to leisure. Examples: tax sporting equipment, household appliances, recreational vehicles, etc.

- This indirectly lowers demand for leisure and acts like a tax on leisure
- Gets us a bit closer to the **most efficient outcome**

# Equity in Taxation

- Ramsey rule implies taxing goods that are **inelastic** more heavily
- Makes sense when those goods are socially undesirable (e.g., cigarettes, alcohol)
- But what if those goods are **necessities** (e.g., food, clothing, housing, medicine)?
- Problematic for a few reasons, but one is that it lacks **vertical equity**
  - **Vertical equity**: people with greater ability to pay should pay more in taxes
- You can modify the Ramsey rule to account for vertical equity
  - A modified rule may tax necessities less heavily
  - Essentially allows for **larger excess burden** to achieve **greater equity**

## Think About It

Suppose a government must choose between two tax systems: one that minimizes excess burden (following the Ramsey Rule strictly) and one that is more equitable but generates a larger excess burden. How would you decide how much efficiency to sacrifice for equity? Is there a principle or framework that could guide this tradeoff?

# Summary

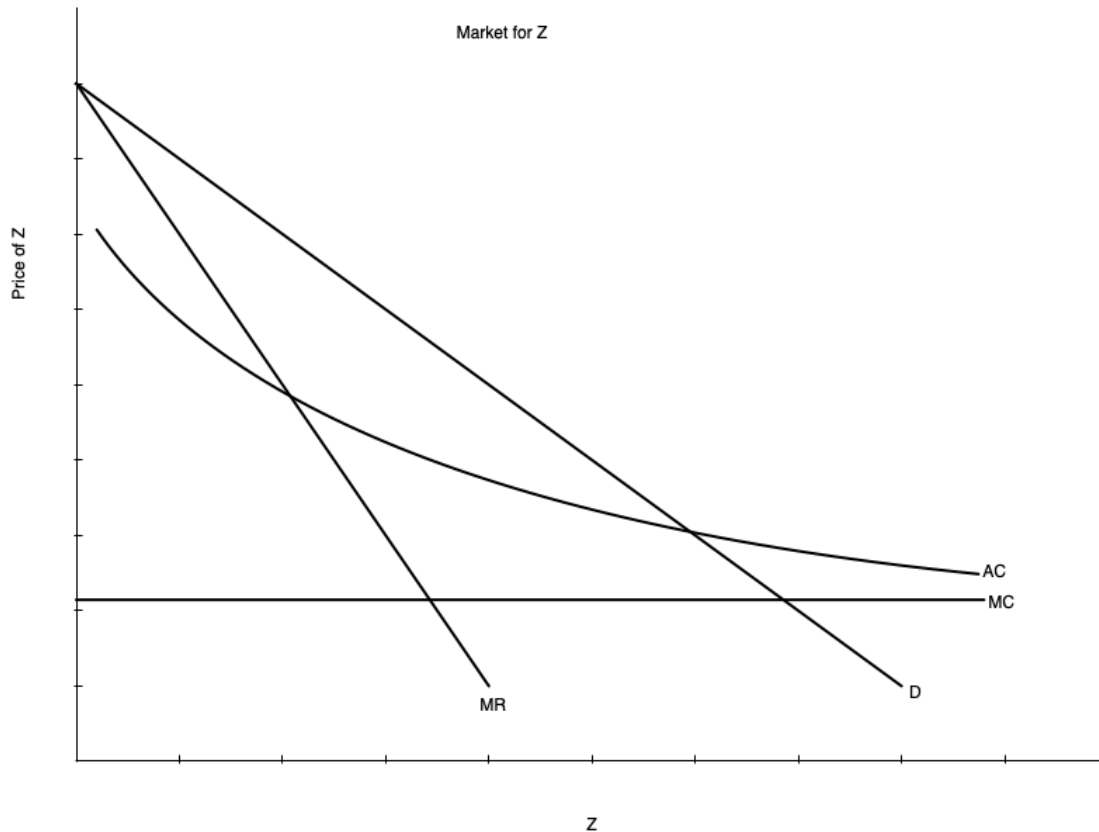
- In the real world we cannot achieve the efficient outcome of a **lump-sum tax** to raise government revenue
- Distortionary taxes change behaviour and create an **excess burden**
- We can design the tax system to minimize the excess burden
- The **Ramsey Rule** gives that optimal design
- But tax efficiency is not the sole consideration
  - **Equity** is also important
  - The optimal rule can change to account for equity considerations

# Optimal User Fees

# Introduction

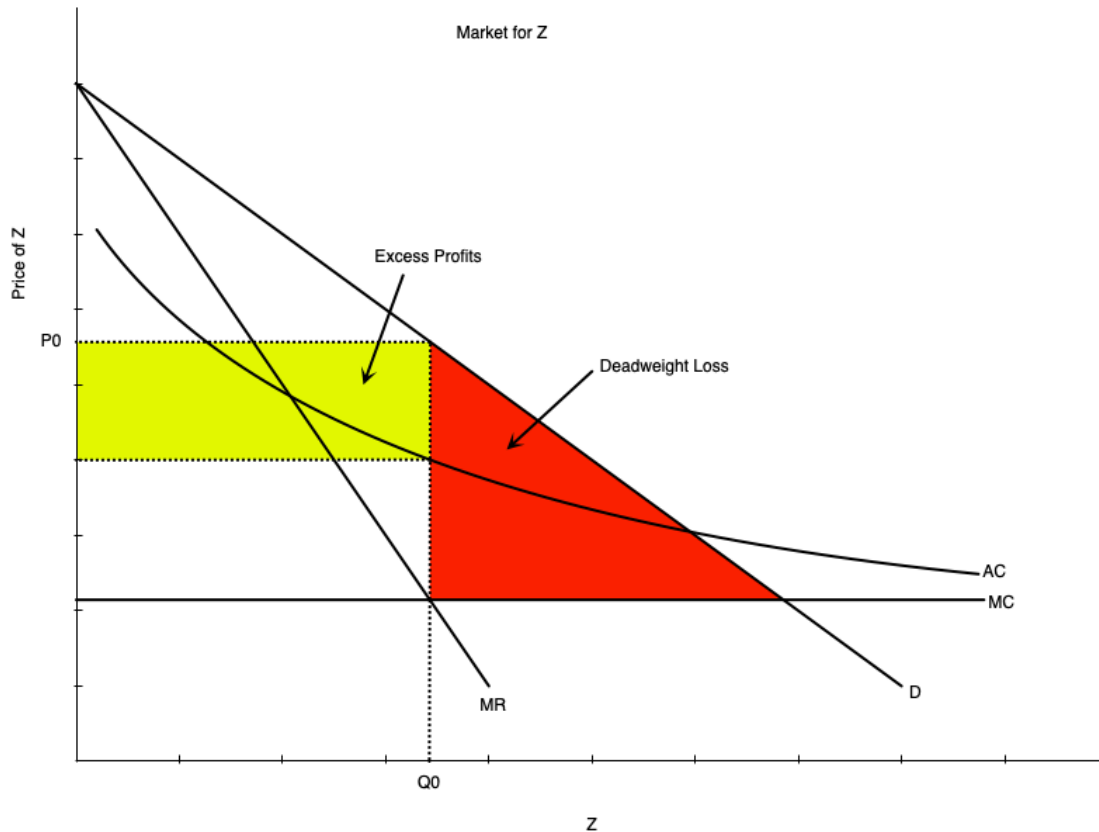
- Governments often charge **user fees** for public services
  - **User fee:** a fee charged by the government for the use of a good or service
  - Examples: park entrance fees, toll roads, public transit fares, etc.
- Like a tax, but not a tax
- Determining the optimal user fee is similar to determining the optimal tax

# Government Production



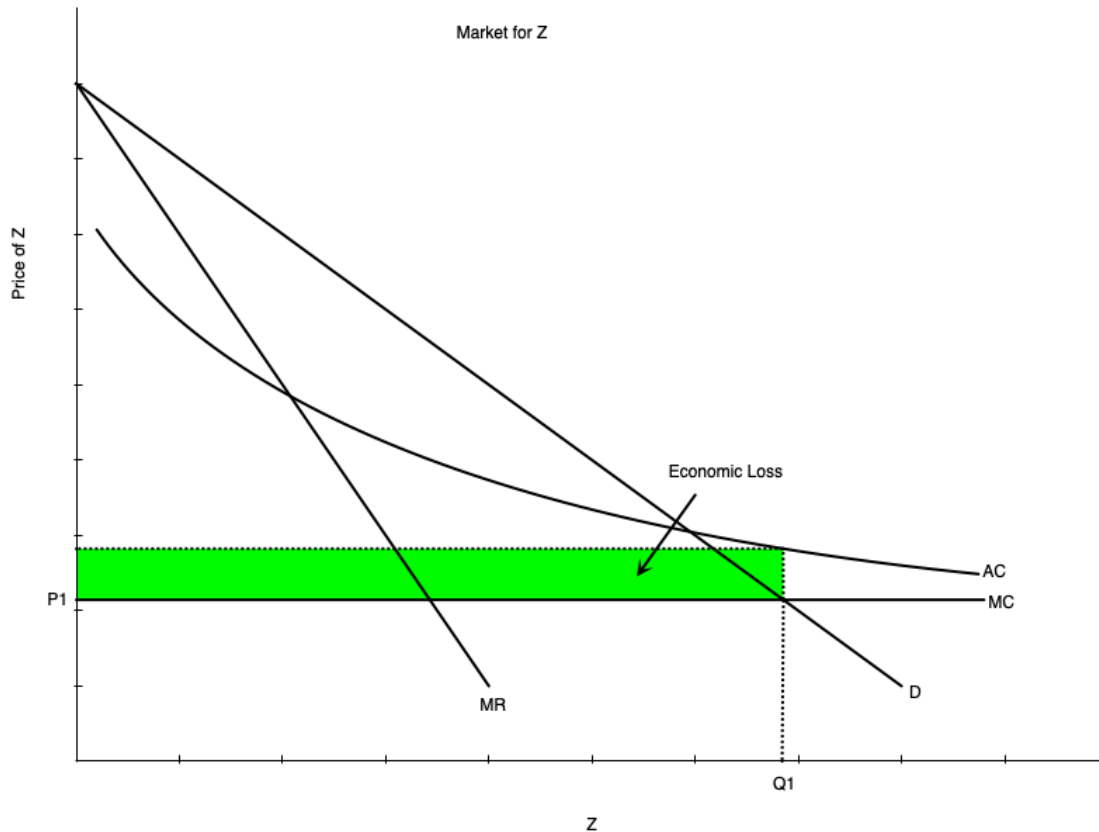
- Governments sometimes produce goods/services when there is a **natural monopoly**
  - **Natural Monopoly:** a market where a single firm can produce the entire output at lower cost than multiple firms
  - Happens with continuously decreasing average costs
  - Examples: water, electricity, public transit, etc.
- Natural monopoly depicted to the left
- Marginal cost fixed for ease of analysis
  - Could also draw it downward sloping as in the text

# Government Production



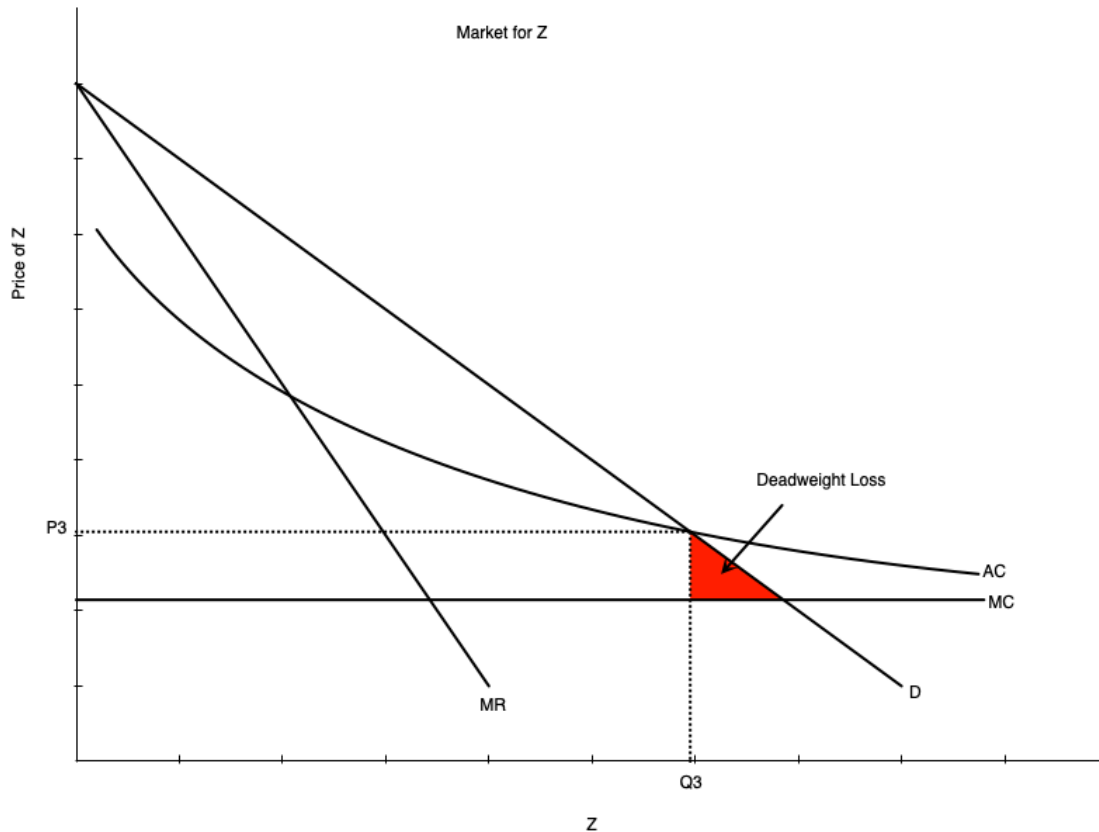
- An unregulated monopolist would produce  $Q_0$  where  $MR = MC$
- Would charge price  $P_0$
- The monopoly earns **excess profits** equal to yellow area
- Also associated with **deadweight loss** (excess burden) equal to red area
- The monopolist does not produce the **efficient** ( $P=MC$ ) level of Z

# Government Production



- The government could take over and produce the good/service
  - Or regulate the private monopolist
- There are different options for production
- One is to produce the **efficient level**  $Q_1$  where  $P = MC$
- But price  $P_1$  is below average cost  $AC$
- The government would incur a **loss** equal to green area

# Government Production



- Another option is **average cost pricing**
- Produce  $Q_2$  where  $P = AC$
- Government incurs **no excess profits or losses**
- Price  $P_2$  is above  $MC$ , so there is still some **deadweight loss** (excess burden) equal to red area

# Government Production

- A third option is to set  $P=MC$  and charge a lump sum tax to cover the loss
  - Hard to do in practice because generally cannot levy lump sum taxes
  - Consumers not using the good would pay for it
- A fourth option is a **two-part tariff**
  - Set  $P=MC$  and charge a fixed fee to cover the loss
  - Only users of the good pay the fee
  - Divide the fee by the number of users to get the fee per user

# Government Production

- Lastly, use the [Ramsey Rule](#)
  - If government produces multiple goods/services, set user fees for each
  - Optimally set them so that the percent reduction in quantity demanded is the same for all goods/services
  - We saw that this involves setting higher fees for goods/services with [inelastic demand](#)
  - The user fee effectively acts like a tax on the good/service

## Think About It

Consider public transit in a city. Marginal cost pricing would set fares very low (since one extra rider costs almost nothing), but transit systems have large fixed costs. If you were advising a city government, which pricing strategy would you recommend – marginal cost pricing with a subsidy, average cost pricing, or a two-part tariff? What are the equity implications of each approach for low-income riders?

# Optimal Income Tax

# Introduction

- So far we have discussed optimal commodity taxes and user fees
- **Income taxes** are another major source of government revenue
- Income taxes are different because they tax a person's **ability to pay**
  - Ability to pay is determined by income
  - Income taxes therefore affect both **labour supply** and **consumption**
- You can design an income tax optimally

# Old School Optimal Tax

- A very old approach to optimal income taxation is [Edgeworth's Model](#)

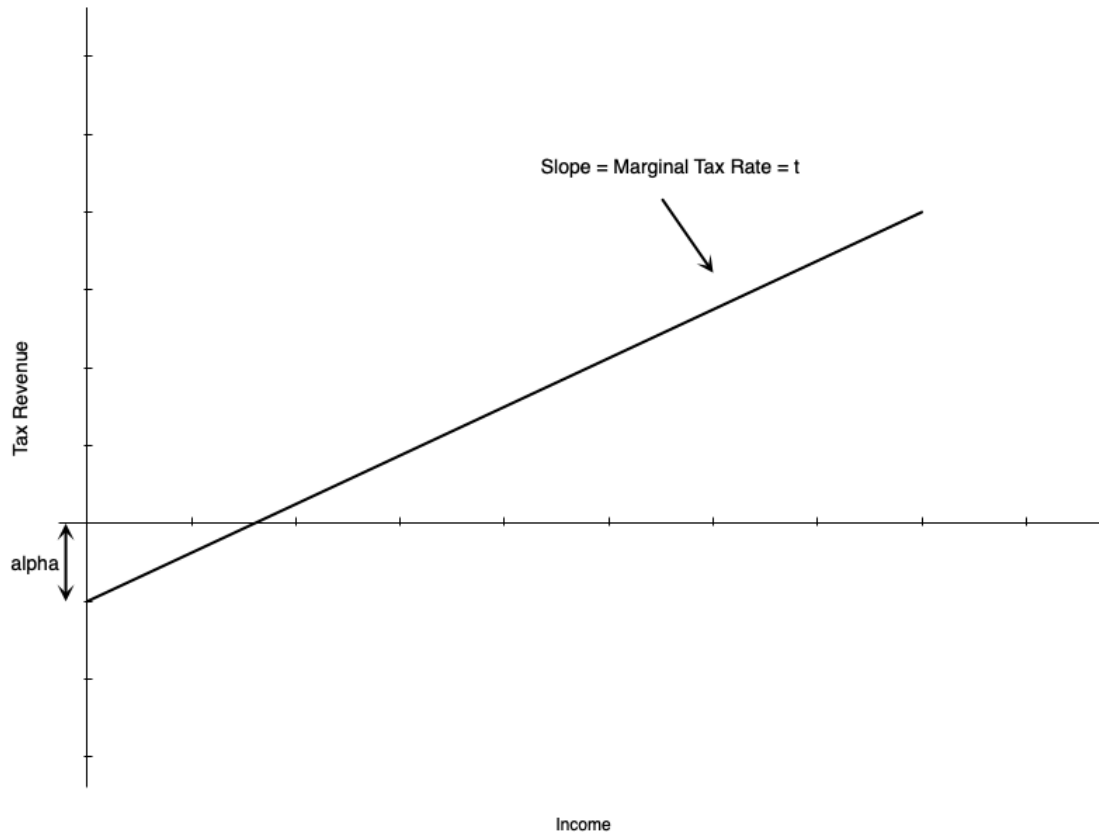
## Edgeworth's Model Assumptions

- Maximize [social welfare](#): the combination of individual utilities
  - Social welfare is [utilitarian](#): the sum of individual utilities
  - Each person's utility is the same, depends only on income
  - Has [diminishing marginal utility](#) of income
  - Total amount of income in society is fixed
- 
- This model predicts that after-tax income is the same for everyone
    - Involves taxing rich and giving to poor until equality is reached
  - This is [not a realistic model](#), but it is a starting point

# Newer Approach to Optimal Tax

- Major problem with Edgeworth's model is that it **ignores behavioural responses**
  - People will supply less labour if taxed heavily
- We know from previous sections that taxes affect labour supply
- Modern models take this into account
  - Taxes raise **revenue**
  - But also create **excess burden** by distorting labour supply
  - Optimal tax balances these two effects
- These models compute optimal tax in a world where people choose labour and leisure

# Newer Approach to Optimal Tax

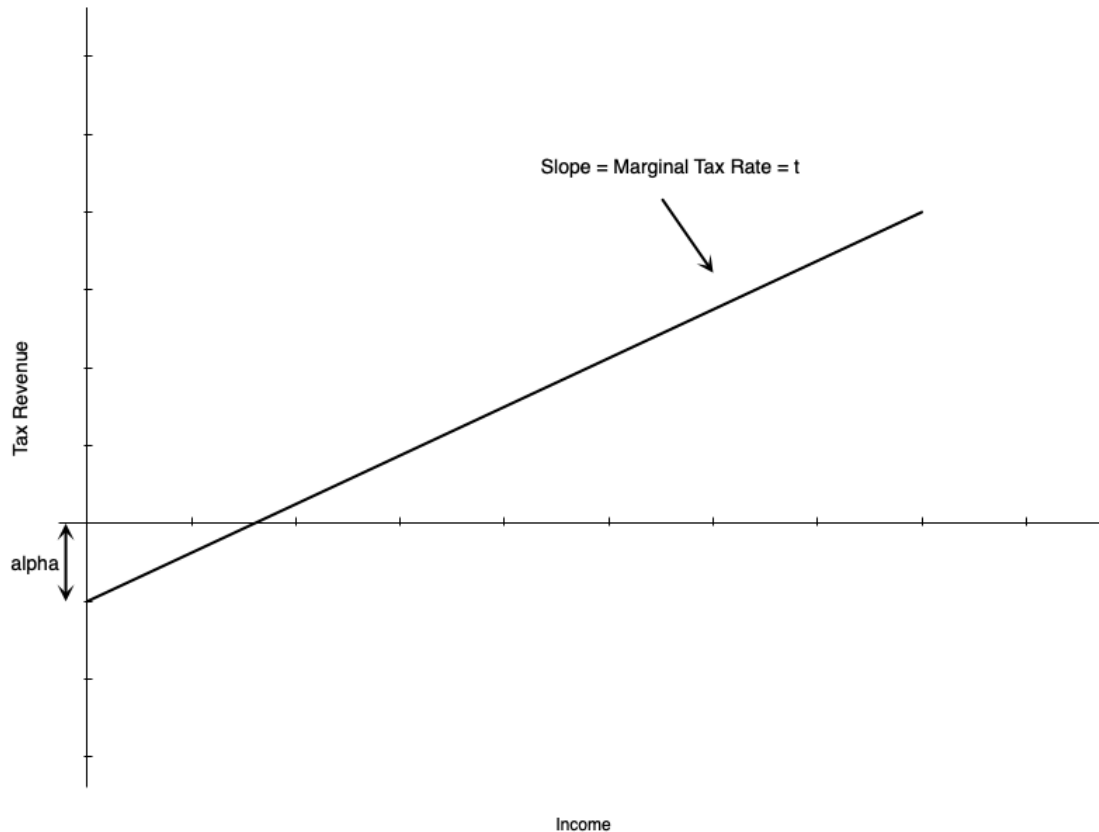


- A simplified version of this model is depicted to the left
- In this setup revenue is

$$Revenue = -\alpha + t \times Income$$

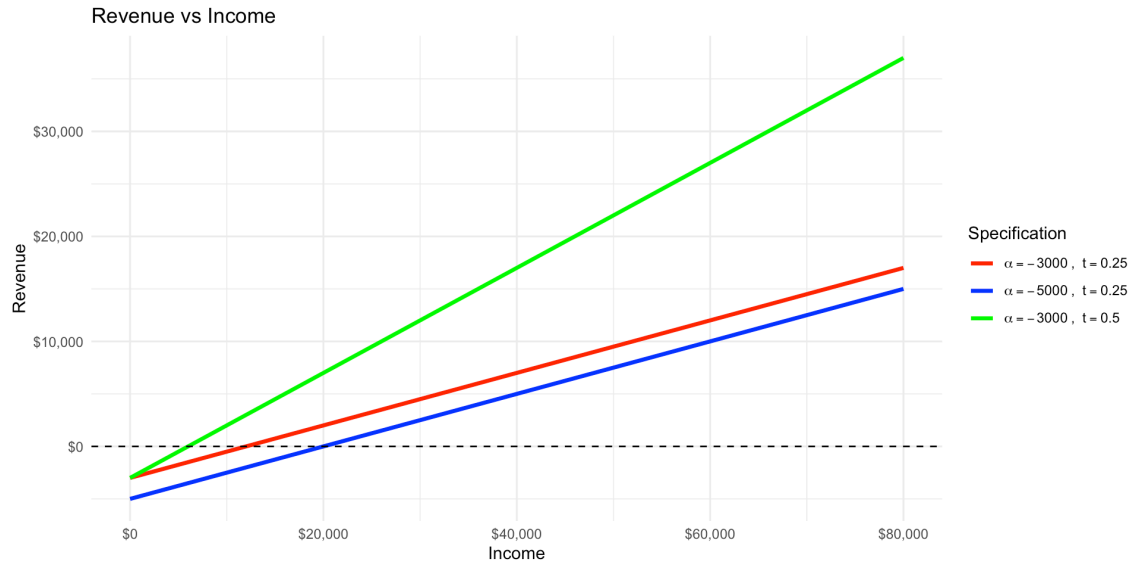
- The term  $-\alpha$  is a fixed payment to everyone
  - If  $\alpha > 0$ , it is a **subsidy**
  - If  $\alpha < 0$ , it is a **lump-sum tax**
- The term  $t$  is the marginal income tax

# Newer Approach to Optimal Tax



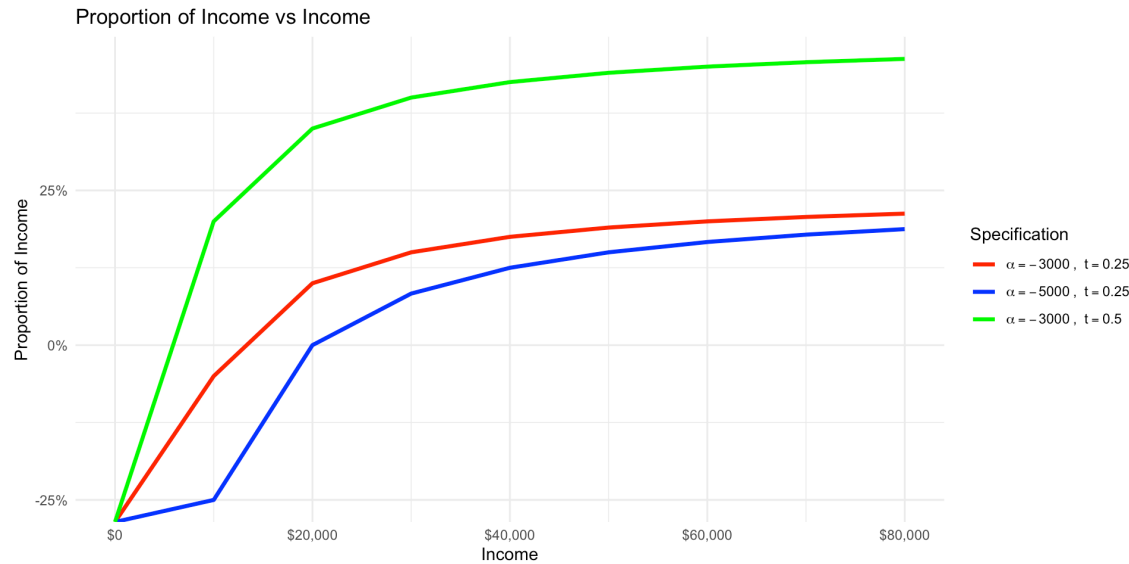
- A constant marginal tax is a **flat tax**
  - Comes from a linear tax schedule
- Studies find that the optimal income tax is **non-linear**
  - Marginal tax rate rises with income
  - Implies a **progressive tax system**
  - This exists in many countries, including Canada and the US
- However, they also find that the optimal non-linear tax is approximated a flat tax
  - So we can still gain insights from the flat tax

# Newer Approach to Optimal Tax



- An example with different flat taxes is shown to the left
- Graph plots total revenue
- There are three curves
  - $\alpha = -3000, t = 0.25$
  - $\alpha = -5000, t = 0.25$
  - $\alpha = -3000, t = 0.5$
- $\alpha$  changes the intercept
- $t$  changes the slope

# Newer Approach to Optimal Tax



- Now plot the proportion of income paid in tax
- Here you can see the **progressivity** of the tax
- The tax with the highest marginal tax rate ( $t = 0.5$ ) and the lowest fixed payment ( $\alpha = -3000$ ) is the most progressive
- The higher the marginal tax rate, the **more progressive** the tax
- The lower the fixed payment, the **more progressive** the tax

# Newer Approach to Optimal Tax

- The specific optimal tax rates depend on the model assumptions
  - Labour supply elasticity
  - Social welfare function
  - Ability to pay
  - Income distribution
- With the utilitarian social welfare function, optimal is around 50%
  - Optimal grant  $\alpha$  is about 60% of average worker income
- Using a more egalitarian social welfare function, optimal tax is higher
  - Egalitarian social welfare function puts more weight on poorer individuals
  - Optimal rate around 80%
- Finally, some studies find that if tax rates are allowed to be different, rich should pay lower marginal rate

# Politics and Time Inconsistency

# Introduction

- **Optimal tax theory** provides purely theoretical guidance on how to design taxes
- In practice, tax policy is determined by **politics**
  - Politicians care about getting elected
  - Voters care about their own taxes and benefits
- So the optimal tax is **not a realistic outcome**
- It is possible that in the real world, implementing the optimal tax is not even desirable

# Time Inconsistency Problem

- Imagine a government wants to tax society to raise revenue
- There are three possible goods that people can consume
  - $X$ ,  $Y$ , and Leisure  $L$
- Labour has a fixed supply so income is also fixed
- The government currently taxes only  $X$ , not  $Y$
- A tax economist suggests lowering the tax on  $X$  and taxing both  $X$  and  $Y$  at the same rate
  - **Efficient** because relative prices are unchanged, no change in labour
  - No excess burden

# Time Inconsistency Problem

- Citizens are cynical
  - They think the government will implement the tax on  $Y$  but **not lower it on  $X$**
  - View the government as trying to maximize revenue at their expense
  - In certain situations they might be right
- If they are correct, taxing only  $X$  is more efficient than the alternative of taxing both

## Time Inconsistency Problem

Optimal tax policy is **not credible** over time. The government cannot commit to future policies. To fully implement optimal tax, the government must credibly commit to future policies – this is difficult in practice, so optimal tax may not be achievable or even desirable.

## Think About It

Can you think of a real-world example where a government introduced a tax that was supposed to be temporary or offset by other changes, but ended up becoming permanent or expanding? What mechanisms could be put in place to help governments credibly commit to tax policy promises?

# Other Criteria for Tax Design

# Horizontal Equity

## ! Horizontal Equity

People in “equal positions” should be treated equally. When it comes to taxation, equal positions might mean equal ability to pay, out of income, consumption, or wealth.

- Horizontal equity is desirable because it is viewed as **fair**
  - People in equal positions should pay the same tax
  - If not, people may view the tax system as **unfair** and try to avoid taxes
- Defining horizontal equity in terms of income or wage is **problematic**
  - Two people who earn the same wage but work different hours have different abilities to pay
  - Taxing wage instead is a problem because it is based on human capital investments

# Horizontal Equity

- Could instead define horizontal equity in terms of **utility**
  - **Utility definition of horizontal equity:** people with the same pre-tax utility should have the same after-tax utility, and taxes should not alter the utility ordering
- Several problems with this approach
  - Utility is **unobservable**
  - If taxing only income, it can penalize people who
    - Consume goods that are income intensive
    - Work in jobs where the pay is purely monetary instead of non-income benefits (e.g. nice work environment, flexible work)
  - Depends on utility ordering prior to any new taxes being imposed (i.e. biased towards status quo)
- Basic idea is that while horizontal equity is desirable, it is **hard to define and implement** in practice



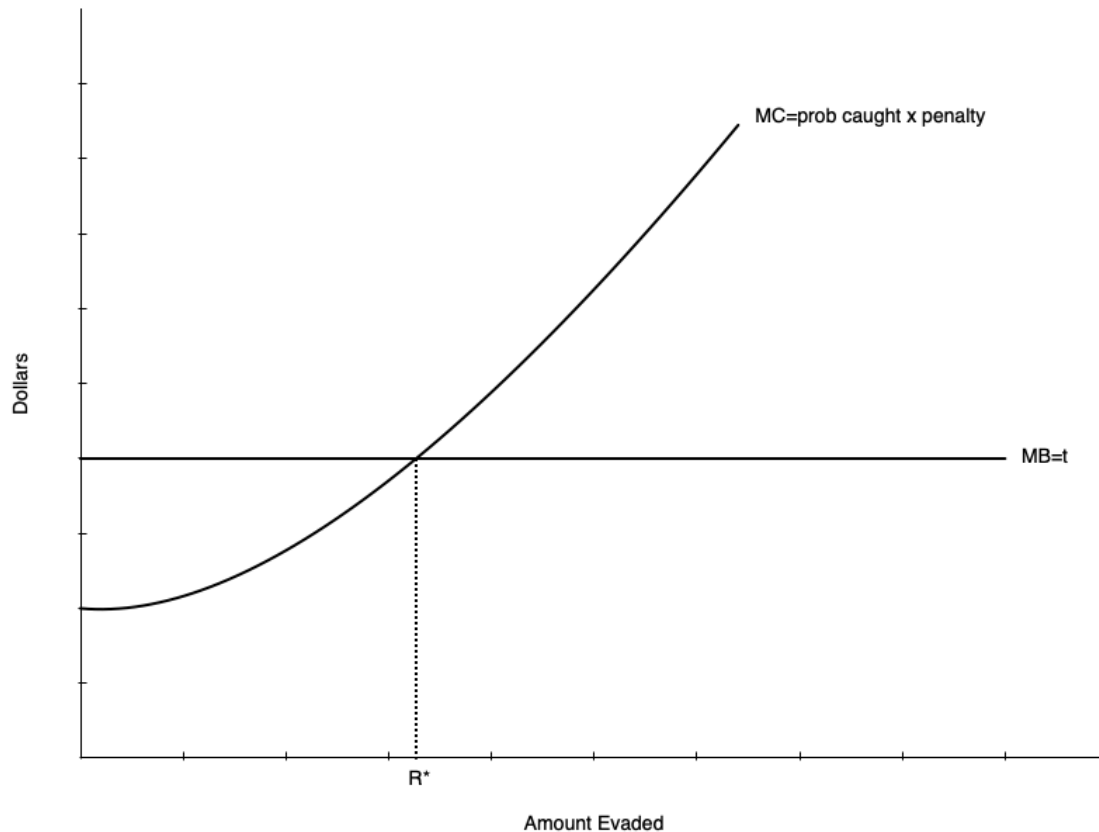
# Administrative Costs

- Taxes are **costly to administer**
  - Direct costs of about \$1 to collect \$100 of tax revenue in Canada
  - Average taxpayer pays about \$200-\$250 in tax preparation costs per year
  - Overall it costs about **2% of GDP** to collect taxes in Canada
- Tax design must balance revenue benefits against administrative costs
- Some changes have led to **smaller costs**
  - Online filing, automatic download of tax forms
  - Potential for future: automatic filing

# Compliance

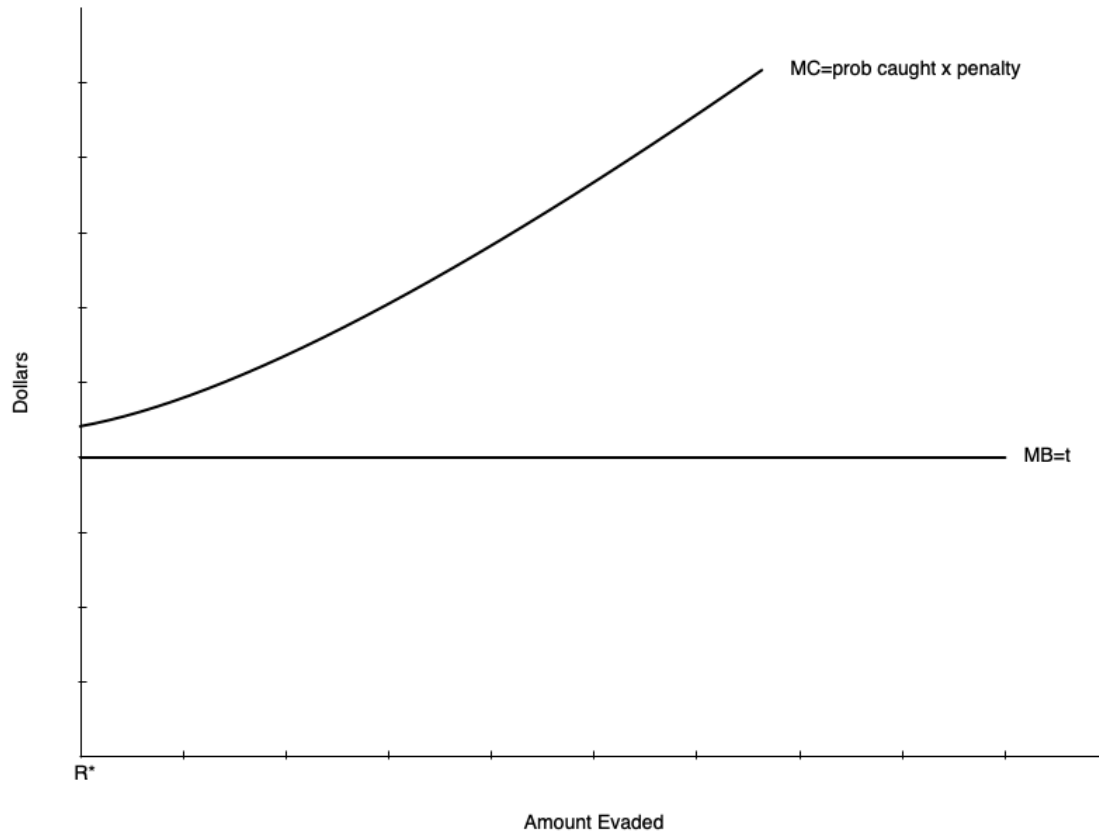
- Canada and other countries operate on a [self-assessment system](#)
  - Taxpayers are responsible for reporting their income and calculating their taxes
  - Government audits a small percentage of taxpayers to ensure compliance
- Some people engage in **tax avoidance**
  - **Legal ways**: changing behaviour to minimize taxes
  - **Illegal ways**: tax evasion, not reporting income, fraud
- It is not always easy to catch tax evasion
  - Government has to balance the costs of enforcement against the benefits
  - If enforcement is too costly, it may not be worth it
- There is an economic theory that determines “optimal” tax evasion

# Compliance



- Imagine that costs of evasion rise with the amount evaded
  - Penalty rises with amount evaded
  - There is some probability that you get caught
  - Could be psychic costs to cheating
  - Risk aversion might play a role
- Benefit is constant
  - Gain is  $t$  per \$1 evaded
- As with many economic decisions, optimal evasion occurs where **marginal benefit equals marginal cost**
- This person would evade  $R^*$

# Compliance



- If costs are too high, **no evasion** occurs
  - Probability of getting caught might be high
  - Penalty might be high
  - Large psychic costs or very risk averse

# Compliance

- Also the normative question about tax evasion: do we care?
- Some economies have large **underground or informal economies**
  - Developing countries often have significant employment in informal economies
- While it is incorrect to call them tax evaders, people in informal employment do not pay taxes
- It might be desirable to have the underground/informal economy even if they pay no tax
  - They are often **poor** and would not pay much tax
  - They often provide goods/services that are **beneficial to society**
  - Trying to eliminate the informal economy might do **more harm than good**

# Summary

# Summary

- Taxes create **excess burden** by distorting behaviour
- The **Ramsey Rule** gives the optimal way to tax goods to minimize excess burden
- **User fees** can be optimized in a similar manner
- Optimal income tax is **progressive**, but the specific rates depend on model assumptions
- **Politics** and **time inconsistency** can prevent optimal tax from being implemented
- Other criteria for tax design include **horizontal equity**, **administrative costs**, and **compliance**

# References

# References

- Rosen, Harvey S., and Lindsay M. Tedds, and Trevor Tombe, and Jean-Francois Wen, and Tracy Snoddon. Public Finance in Canada. 6th Canadian edition. McGraw-Hill Ryerson, 2023.
- Gruber, Jonathan. Public Finance and Public Policy. 7th edition. Worth Publishers, 2022.